

Math 164 Workshop Problem

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1 Problem

(a) The earth is modelled by the formula $x^2 + y^2 + z^2 = 9$. Find a path of a rocket ship traveling through space such that it only touches the earth surface at the point $(1, 1, \sqrt{7})$. Assume that the rocket can be treated as a point in space.

Solution: Observe that the surface is a sphere. The only possible path that can intersect the surface at one point is if and only if the path is tangent to the surface. There are infinitely many tangent paths at this point so let's just construct one. We must represent the surface as a two-variable function (because we eventually want to find a function that describes the change of $z = F(x, y)$ with respect to x). We rearrange the equation of the surface to

$$z = F(x, y) = \pm\sqrt{9 - x^2 - y^2}$$

Observe that the point $(1, 1, \sqrt{7})$ is in the upper hemisphere so we can write $F(x, y) = \sqrt{9 - x^2 - y^2}$. Next we find any tangent vector. To do so, we will use a simple case by finding F_x where it is given by

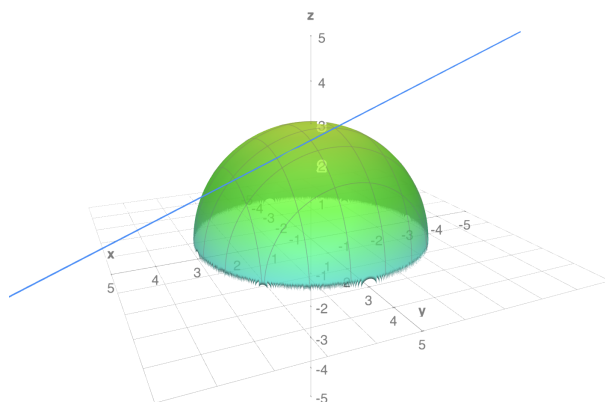
$$F_x = \frac{1}{2}(9 - x^2 - y^2)^{-\frac{1}{2}}(-2x)$$

Since we constructed a function dependent on only x and y , we can simplify the point $(1, 1, \sqrt{7})$ in three-variables to $(1, 1)$ in two-variables. We find $F_x(1, 1)$ as

$$F_x(1, 1) = \frac{1}{2}(9 - 1^2 - 1^2)^{-\frac{1}{2}}(-2 \cdot 1) = -\frac{1}{\sqrt{7}}$$

We must convert this scalar value into a vector. Observe that $F_x = -\frac{1}{\sqrt{7}}$ represents the change of $F(x, y) = z$ with respect to x . You can think of it as "for each unit of x , z decreases by $\frac{1}{\sqrt{7}}$ units." So, we can represent this by the vector $(1, 0, -\frac{1}{\sqrt{7}})$. This vector is the directional vector of the path of the rocket. Thus we have a parametric equation for the path of the rocket, let's call it L

$$L = \begin{cases} x = 1 + t \\ y = 1 \\ z = \sqrt{7} + 2t \end{cases}$$



Our findings from part (a) graphed. The upper half of the surface with the parametric line

(b) Create an object representation of all possible paths of the rocket ship so that it only touches the earth's surface at the point $(1, 1, \sqrt{7})$.

Solution: From the previous problem we mentioned that there are infinitely many different paths for the rocket to take so that it only intersect that point once. We could represent all possible paths for the rocket to take by constructing a plane tangent to the surface at the point. Constructing a plane is much simpler than a line. For convenience we will define a three-variable function $F(x, y, z) = x^2 + y^2 + z^2 = 9$. Since we are dealing with a three-variable function, the equation for the plane will be represented by

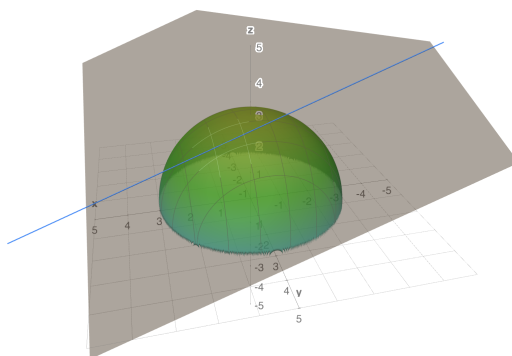
$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

We have $F_x = 2x$, $F_y = 2y$, $F_z = 2z$. So our equation of the plane is

$$2(x - 1) + 2(y - 1) + 2\sqrt{7}(z - \sqrt{7}) = 0$$

Cleaning it up a bit we have

$$2x + 2y + 2\sqrt{7}z - 18 = 0$$



Our findings from part (b) graphed with the line from part (a). Notice how the line is contained in the plane